A search for ultraviolet laser lines in archival spectra of nearby stars

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Abstract

A laser aimed at Earth from another star adds an unresolved emission line to the star's spectrum. We search the ultraviolet, where a diffraction-limited beam fills a solid angle twenty times smaller than at visible wavelengths and cool stars are faint: 2356 HST/STIS echelle exposures of 530 stars from StarCAT and 4312 taken since, of 881 stars, 1138 HST/COS far-ultraviolet spectra of 44 stars within 10 pc, 851 IUE high-dispersion images of 58 stars within 10 pc and 70 FUSE observations of 42 stars within 20 pc. A candidate is an unresolved peak above the local upper envelope of the spectrum that recurs in a second exposure, grating offset or detector channel and that fewer than two other stars show at its rest or its observed wavelength, whether detected there or measured there directly. No candidate remains. The features that persist in one star are narrow windows in the line-blanketed spectrum of Arcturus at the wavelengths of its iron-group ions, emission lines of the Mira R Leo, the symbiotic EG And and ϵ Eri, lines of the crowded ultraviolet spectra of cool giants, T Tauri stars, the Ap star GY And and the chromospheres of α Cen A and B and Procyon, IUE features fixed to one aperture's position on the camera, and IUE features 440 km s^{-1} redward of the Si II 1808 and 1817 Å lines of the brightest G and K dwarfs, which the STIS spectrum of α Cen A excludes. Lines injected into the archival spectra and recovered by the same detectors give the sensitivity. At the stars within 10 pc the median limit for an isotropic source is 2.0×10^{17} W with COS and 1.0×10^{18} W with the later STIS spectra, and a transmitter aimed at Earth through a diffraction-limited 10 m aperture, whose beam at 1200 Å is 0.005 au wide at the nearest stars, would have been detected at 0.1 to 7 W from 21 of the 27 stars within 5 pc, at 0.11 W from Lalande 21185 and 0.14 W from Barnard's star.

Keywords: Technosignatures, Lasers, Ultraviolet: stars, Techniques: spectroscopic, Solar neighbourhood

1. Introduction

Schwartz and Townes proposed that a civilization could signal with optical masers, whose light, confined to a narrow beam and a narrow band, would outshine its star at that wavelength [1]. A laser aimed at Earth from another star would add an unresolved emission line to the star's spectrum, and spectroscopic searches for such lines have examined optical spectra of nearby stars: 577 FGKM stars with Keck/HIRES at 4000–5000 Å [2], 2796 stars at 3640–7890 Å [3], 5600 FGKM stars with thresholds of megawatts [4], 1983 spectra of 388 stars from the Automated Planet Finder [5], and 2821 stars in archival HARPS spectra [6].

The ultraviolet favours such a transmitter in two ways. A diffraction-limited beam from an aperture D fills a solid angle of about $(\lambda/D)^2$, so at 1200 Å an aperture puts its power into a solid angle twenty times smaller than at 5500 Å; from 10 m the beam is 0.005 au wide at 2 pc and 0.025 au at 10 pc, far narrower than the Earth's orbit, so the transmitter must be aimed at the

Earth itself, at its position when the light arrives, and not at the Sun. And the photospheres of K and M dwarfs, the commonest nearby stars, emit almost no far-ultraviolet continuum, so a line of given power stands out against far less light. The cost is that the atmosphere absorbs the ultraviolet, so only a receiver in space sees the signal, and that the interstellar medium absorbs the Lyman series and the resonance lines of abundant ions. Within 10 to 20 pc the hydrogen column is of order 10^{18} cm^{-2} [8], which leaves the medium transparent longward of 912 Å outside those lines.

This paper asks whether archival ultraviolet spectra of nearby stars hold such lines. Four archives hold high-resolution ultraviolet spectra of them: those of the Space Telescope Imaging Spectrograph (STIS) [9], whose echelle spectra of 545 stars are collected in StarCAT [10]; of the Cosmic Origins Spectrograph (COS) [11]; of the International Ultraviolet Explorer (IUE) [12], in its final processing [13]; and of the Far Ultraviolet Spectroscopic Explorer (FUSE) [14, 15, 16]. We search all four for unresolved emission lines with one set of criteria, and measure what each search could have found by passing lines injected into the archival spectra through the same detectors.

1.1. Contributions

1. *A search of four archives.* 2356 STIS echelle exposures of 530 stars from StarCAT and 4312 later exposures of 881 stars from the pipeline, 1138 COS spectra of the 44 stars within 10 pc that COS observed, 851 IUE high-dispersion images of the 58 stars within 10 pc that IUE observed, and 70 FUSE observations of 42 stars within 20 pc, each spectrum credited to the star in the aperture (Section 2). A candidate must be unresolved, rise above the local upper envelope of the spectrum, recur in a second exposure, grating offset or detector channel, and be absent from other stars at its rest and its observed wavelength (Section 3).
2. *What the tests remove.* No candidate remains (Table 3). The features that persist in one star and that no two other stars share are windows between the absorption lines of Arcturus at the wavelengths of its iron-group ions, Fe II and C III] emission of the Mira R Leo and the symbiotic EG And, chromospheric lines of ϵ Eri (Table 4), and in the later STIS spectra 138 lines of eleven stars whose ultraviolet spectra are crowded with natural lines. In IUE spectra the tests remove features fixed to one aperture's position on the camera, and features 440 km s^{-1} redward of Si II 1808 and 1817 Å in the brightest G and K dwarfs, which other stars show below the detection criteria and the STIS spectrum of α Cen A excludes (Section 4.1).
3. *Sensitivity.* Injected lines are recovered at 90% of each search's plateau at 2.8 to 4.7 times the nominal threshold. At the stars within 10 pc the median limit for an isotropic source is 2.0×10^{17} W with COS, 1.0×10^{18} W with the later STIS spectra and 3.9×10^{18} to 2.9×10^{21} W with the others, and from 21 of the 27 stars within 5 pc a transmitter aimed at Earth through a 10 m aperture would have been detected at 0.1 to 7 W (Section 4.2, Table 5).

Section 2 describes the data and how each spectrum is credited to a star, Section 3 the detection, the tests and the measurement of sensitivity, and Section 4 the candidates and the limits. Section 5 discusses related work and Section 6 concludes.

2. Data

Table 1 lists the data. From StarCAT we use every single-exposure spectrum in the echelle settings E140M, E140H, E230M and E230H, 2356 exposures of 530 objects observed from 1997 to 2004, with the flux, error and data-quality arrays and heliocentric vacuum wavelengths of its

Table 1: The ultraviolet spectra searched. Spectra are single exposures (STIS, from StarCAT and, later, from the pipeline's x1d files), segment spectra (COS), echelle images (IUE) and observations of eight channels (FUSE). The COS and IUE stars are those within 10 pc, the FUSE stars those within 20 pc, and StarCAT holds stars at all distances.

Instrument	Settings	Range (Å)	R	Years	Spectra	Stars
STIS	E140M, E140H, E230M, E230H	1150–3100	30,000–114,000	1997–2004	2356	530
STIS, later	E140M, E140H, E230M, E230H	1150–3100	30,000–114,000	2004–2026	4312	881
COS	G130M, G160M	1065–1800	16,000–24,000	2009–2026	1138	44
	G140L	1100–2150	2,000–3,000			
IUE	SWP, LWP, LWR	1150–3350	10,000	1978–1995	851	58
FUSE	LiF, SiC	905–1187	17,000	1999–2007	70	42

processing. For the STIS exposures taken after StarCAT, from 2004 July on, we take the pipeline's extracted x1d spectra of every echelle exposure in the same four settings: 4402 files, of which the archive served 4312, of 881 stars observed to 2026; the 90 it did not serve belong mostly to AU Mic and ξ^1 CMa. For COS we build spectra from the photon events of every far-ultraviolet exposure of a star within 10 pc: events with good data-quality flags and pulse heights, outside the flashes of the wavelength-calibration lamp, in bins of two pixels along the dispersion, with the background scaled from the pipeline's background regions and the heliocentric wavelengths of the pipeline, leaving out the lamp, dark and flat exposures filed under science visits. This gives 1138 segment spectra of 44 stars from 2009 to 2026. For IUE we use the net flux and noise of every NEWSIPS high-dispersion image of a star within 10 pc at the pixels with quality flag zero, 851 images of 58 stars from the short-wavelength (SWP) and long-wavelength (LWP, LWR) cameras, 1978 to 1995. For FUSE we use the combined spectra of the eight channels, LiF and SiC on the detector segments 1A, 1B, 2A and 2B, which record each wavelength two or three times on different detectors, of every observation of a star within 20 pc: 70 observations of 42 stars, 1999 to 2007.

A star lies within 10 pc (20 pc) when its SIMBAD parallax [17] exceeds 100 mas (50 mas). The coordinates an archive gives for an observation are the observer's catalogue position of the target, of any epoch between 1950 and the observation, so a pointing is matched to the stars whose proper-motion track over that interval passes near it, and a spectrum is credited to the nearest such star when the track lies inside the aperture: within 3″ for COS, whose aperture is 2.5″ across, within 3″ for the later STIS exposures, or 15″ when the observer's target name resolves to that star, since the orbital motion of a binary such as α Cen carries its stars several arcseconds off their catalogue tracks, and within 15″ for FUSE, whose large aperture is 30″ square. Some archived IUE coordinates are 30″ in error, so an IUE image is credited to the star its target name resolves to in SIMBAD, and otherwise to the nearest track within 10″. FUSE observations that place the star at the reference point, outside every aperture, are left out. A spectrum of a companion is thereby credited to the companion, and pointings at fast-moving stars such as Barnard's star, which moves 10″ a year, are matched.

3. Method

3.1. Detection

Let f_i and σ_i be the flux and its error at pixel i, and e_i the 90th percentile of the flux in a running window. Between dense absorption lines a running median falls below the continuum and

Table 2: Detection and persistence. The envelope is the running 90th percentile of the flux over the window given; a persistent feature recurs within a third of a resolution element (COS: two pixels). IUE is searched twice, at 6σ in two images and at 4.5σ in three.

	Envelope	Peak	Persistence
STIS	1 Å	$z \geq 6$, 3σ above the rest of ± 2.5 Å	two exposures with one grating
COS	2 Å, sums of six pixels	Poisson $p < 10^{-9}$	two exposures at two FP-POS settings
IUE	5 Å	$z \geq 6$ (4.5), 3σ above the rest of ± 2.5 Å	two (three) images with one camera and aperture
FUSE	2 Å	matched filter, 5σ	3σ in a second channel within $20\,\mathrm{km\,s^{-1}}$

the gaps between lines rise above it, so a peak is measured against this upper envelope,

$$z_i = \frac{f_i - e_i}{\sigma_i}.$$

For STIS, in both sets, and IUE a peak must reach $z_i \geq 6$ and exceed every other pixel within 2.5 Å by $3\sigma_i$, and a Gaussian fitted to it must have a full width at half maximum between one half and one and a half instrumental widths λ/R, centred within one of them. For COS the counts are summed over three bins, six pixels, and a peak must have a Poisson probability below 10^{-9} against the larger of the envelope of these sums and the background, with a fitted width of 3.3 to 9.8 pixels, half to one and a half times the 6.5 pixels of the line spread function. For FUSE a filter matched to a Gaussian of $\sigma = 0.026$ Å, the line spread function of a point source, gives the amplitude of a line at each pixel in units of its error; a peak must reach 5 in one channel, with a fitted σ of 0.013 to 0.052 Å, narrower than the airglow lines that fill the aperture. Table 2 gives the windows and the persistence required of each instrument. The geocoronal lines and the strongest chromospheric and transition-region lines are masked in the STIS, COS and IUE spectra: H I Lyα, O I 1302–1306 and 1356–1358 Å and N I 1200 Å within 0.6 Å (Lyα within 2 Å for COS), and C III 1176, Si III 1206, N V, C II 1335, Si IV, C IV, He II 1640 and Mg II within 1.2 Å.

3.2. Persistence

A laser on a star stays at the star's wavelength, while a defect of the detector stays at one pixel. A feature must recur in a second STIS exposure with the same grating, in a second COS exposure at a different FP-POS setting, which moves the spectrum along the detector, in a second IUE image with the same camera and aperture, or, for FUSE, at 3σ within $20\,\mathrm{km\,s^{-1}}$ in a second channel that records the same wavelength on another detector (Table 2).

3.3. Tests against other stars

A feature that other stars share is a line of their spectra, of the instrument or of the sky. Two tests compare each persistent feature with the other stars. The first counts the other stars with a detected peak within $15\,\mathrm{km\,s^{-1}}$ ($20\,\mathrm{km\,s^{-1}}$ for FUSE) of the feature's rest wavelength, each star in its own rest frame from its SIMBAD radial velocity, and of its observed wavelength; the peaks counted are those at the detection threshold for STIS and IUE, at $p < 10^{-5}$ for COS, and those confirmed in a second channel for FUSE. The second measures without the detection criteria: the largest z_i within two pixels of the feature's rest wavelength, in each other star's frame and in

every spectrum of it from the same instrument (for IUE, the same camera), and of its observed wavelength in every spectrum with the same setting (for STIS, grating and central wavelength; for IUE, camera and aperture). For COS the second test is the sum of all other stars' spectra in the same grating, each in its rest frame, in which a feature with Poisson probability below 10^{-5} is a stellar line. A feature that two or more other stars show, by either test and in either frame, at $z > 5$ in the second, is removed. An IUE feature that one aperture shows and that lies below 2σ in every image of the star through the other aperture belongs to the first aperture's position on the camera, since the two apertures put the same wavelength on different pixels, and is removed as well.

3.4. Sensitivity

The nominal threshold of an exposure is the flux of an unresolved line whose peak sits at the detection threshold,

$$F_{\rm nom} = k\,\sigma_F\sqrt{2\pi}\,\sigma_\lambda, \qquad \sigma_\lambda = \frac{\lambda}{2.355\,R},$$

with $k = 6$ for STIS and IUE, σ_F the flux error per unit wavelength, and the median taken over the exposure's unmasked pixels. For COS it is the number of counts a line needs to reach the Poisson threshold, divided by the 72% of the line spread function that falls within six pixels and converted to flux with the sensitivity of each grating setting from the pipeline's extracted spectra; for FUSE it is the larger of the 5σ flux in the most sensitive channel covering each wavelength and the 3σ flux in the next. Lines of $m = 1$ to 8 times the local nominal flux are injected at random unmasked wavelengths, six per exposure and multiple (for IUE, one per echelle order), into 300 StarCAT and 300 later STIS exposures, 288 COS spectra, 138 IUE images and 69 FUSE observations, and the same detector is run on the result (Figure 1a). Recovery rises to a plateau, the fraction of the band in which a line can be found at all, since a line on a strong stellar or airglow line lies below the envelope. We take as a star's limit the flux at which recovery reaches 90% of the plateau, $m_{90}F_{\rm nom}$, with $F_{\rm nom}$ that of the star's second most sensitive exposure in one setting, since a feature must persist in two; for FUSE, that of its most sensitive observation, whose recovery already requires the second channel. The recovery measures detection in single exposures; the tests against other stars remove a laser only where its wavelength coincides with a feature of two other stars.

A source at distance d that radiates isotropically and gives a line flux F at the Earth has luminosity $L = 4\pi d^2 F$. A transmitter aimed at Earth whose diffraction-limited beam from an aperture D fills a solid angle $(\lambda/D)^2$ needs a power

$$P = F\,d^2\left(\frac{\lambda}{D}\right)^2,$$

which we evaluate for $D = 10\,$m at the median wavelength of the spectra that give each limit; the beam, $\lambda d/D$ across, is 0.005 to 0.025 au wide at the stars within 10 pc, so it must be pointed at the Earth's position at arrival rather than at the Sun, which a transmitter that knows the Earth's orbit can do.

4. Results

4.1. Candidates

Table 3 gives the number of features at each stage and Table 4 the features that pass both tests in one star. The 11 STIS features left lie in three evolved stars. Seven are in the K1.5 giant

Table 3: Features at each stage. Peaks are detections in single exposures (FUSE: in single channels); persistent features recur as Table 2 requires; the first and second tests are those of Section 3.3, the second for IUE including the comparison of apertures and for COS the other stars' summed spectra. Left are the features that pass every test, all identified in the text.

	Peaks	Persistent	First test	Second test	Left
STIS	313	16	13	11	11
STIS, later	4043	318	217	138	138
COS	280	30	21	3	3
IUE, 6σ	643	73	8	0	0
IUE, 4.5σ	1320	61	3	0	0
FUSE	9	0	–	–	0

Table 4: The persistent features that pass both tests against other stars. Wavelengths are in vacuum in the star's rest frame. For STIS, n gives the exposures holding the feature and the exposures in the setting, and Peak the largest z; for COS, n is the number of exposures and Peak the smallest Poisson probability.

Star	Type	Setting	$\lambda_{\rm rest}$ (Å)	n	Peak
Arcturus	K1.5 III	STIS E230H	2638.708	6/13	25.3
Arcturus	K1.5 III	STIS E230H	2700.655	3/13	26.0
Arcturus	K1.5 III	STIS E230H	2746.177	2/13	14.1
Arcturus	K1.5 III	STIS E230H	2765.438	3/13	7.5
Arcturus	K1.5 III	STIS E230H	2765.901	4/13	10.1
Arcturus	K1.5 III	STIS E230H	2978.266	2/13	15.8
Arcturus	K1.5 III	STIS E230H	3072.258	2/13	34.4
EG And	M2.4 III, symbiotic	STIS E230M	1908.816	3/16	7.9
R Leo	M7–9e, Mira	STIS E230M	2612.492	4/4	6.8
R Leo	M7–9e, Mira	STIS E230M	2626.304	4/4	13.4
R Leo	M7–9e, Mira	STIS E230M	2733.136	4/4	8.6
ϵ Eri	K2 V	COS G130M	1330.925	3	$p = 4 \times 10^{-15}$
ϵ Eri	K2 V	COS G160M	1468.405	4	$p = 10^{-18}$
ϵ Eri	K2 V	COS G160M	1769.775	2	$p = 4 \times 10^{-13}$

Arcturus, between 2638 and 3072 Å, and recur in every exposure that covers them with a signal-to-noise ratio above 20, from 1998 to 2003. A laser would recur too, so what identifies them is their relation to the spectrum around them. The near-ultraviolet spectrum of a K giant is blanketed by the lines of the iron-group ions, and between 2600 and 3100 Å the flux reaches the continuum only in narrow windows between them, which the upper envelope cannot follow and the detector reads as unresolved peaks. In every exposure each of the seven sits at the brightest flux within 10 Å of it, within 10 per cent of the next brightest sample for five of them and below it for the other two, as a window does and an emission line above the continuum need not; the same rest wavelengths carry peaks at $z > 5$ in the G dwarfs 61 Vir (2638.7 Å), ξ Boo (2746.2 Å) and in the K giant μ Leo (3072.3 Å), one star each, below the two the second test requires; and their wavelengths are those of the ions whose lines shape the spectrum, at the star's own velocity: V II 2638.711 and Cr II 2638.705 (0.3 and 0.4 km s^{-1}), Mn II 2700.654 (0.1 km s^{-1}), Fe II 2746.177 (0.1 km s^{-1}), Cr II 2765.461 (2.5 km s^{-1}), Fe II 2765.945 (4.8 km s^{-1}), Fe II 2978.219 (4.7 km s^{-1}) and Cr I 3072.303 (4.3 km s^{-1}) in the NIST database [18], where the 4174 lines of these ions with arc intensities of 100 or more over the range put a random wavelength within 0.5 km s^{-1} of one with probability

0.08 and within $5\,\mathrm{km\,s^{-1}}$ with probability 0.8. They are features of the star's spectrum, and the Arcturus ultraviolet atlas [7] is where each can be read against the identified lines around it. Three are in the Mira R Leo, in all four exposures of one visit in December 1998, 50 to 75% above every other pixel within 2.5 Å; the ultraviolet lines of Mira variables other than Mg II are mostly narrow Fe II emission [19], and the strongest of the three lies $3.6\,\mathrm{km\,s^{-1}}$ from Fe II 2626.27 Å. The last is in the symbiotic star EG And, $13\,\mathrm{km\,s^{-1}}$ from the C III] 1908.73 Å intercombination line, and reaches $z > 5$ in seven of the nine exposures that cover it. Two features that pass the first test, in Arcturus at 3070.70 Å and in μ Leo at 3110.08 Å, reach $z > 5$ in two or more other stars and are removed by the second.

The later STIS exposures hold 318 persistent features in 24 stars, of which 217 pass the first test and 138 the second, in eleven stars: 60 in the Ap star GY And, whose spectrum is a forest of rare-earth lines, 52 in Betelgeuse, five each in Procyon and Pollux, four each in α Cen A and B, three in γ Cru, two in TW Hya and one each in β Pic, η Car and α Her. They recur in many exposures of each star, in Betelgeuse in up to 73 of 214, at up to 46σ, and 84 of the 138 lie within $15\,\mathrm{km\,s^{-1}}$ of a line in the NIST database of the ions of stellar chromospheres and transition regions, where a random wavelength does so with probability 0.70, so the database neither confirms nor excludes them: they are lines of spectra so crowded that the upper envelope cannot follow them, the α Cen features among them within $15\,\mathrm{km\,s^{-1}}$ of Fe II and Cr II lines at 6 to 12σ in up to 11 of their 13 exposures. No feature of a star with a clean spectrum survives.

The 30 persistent COS features lie in five stars, 24 of them in ϵ Eri. Nine are detected in two or more other stars and 18 more appear in the other stars' summed spectra. The three left are in ϵ Eri, the young and active K2 dwarf, within $2\,\mathrm{km\,s^{-1}}$ of Cr II 1330.93, C I 1468.41 and Si I 1769.79 Å, where a random wavelength lies within $2\,\mathrm{km\,s^{-1}}$ of one of the NIST lines compared [18] with probability 0.12. They recur in most of the 24, 33 and 14 exposures that cover them from 2015 to 2025, at a significance that follows the depth of each exposure, and the first two appear in the other stars' summed spectra at Poisson probabilities of 6×10^{-5} and 4×10^{-4}. They are lines of the chromosphere of ϵ Eri, which only its deep spectra show.

Of the 73 persistent IUE features at 6σ, eight pass the first test. Four, in small-aperture SWP images of χ^1 Ori, lie below 2σ in all of its large-aperture images. Three, in small-aperture images of ξ Boo B and Procyon, reach $z > 5$ at the same observed wavelength in four or five other stars' images through the same camera and aperture. The eighth, in 5 of the 15 large-aperture SWP images of α Cen A at 1810.64 Å, lies 2.65 Å, $440\,\mathrm{km\,s^{-1}}$, redward of Si II 1808.01 Å; the 4.5σ search adds one at 1819.58 Å, at the same offset from Si II 1816.93 Å. The same offset from Si II 1808 Å reaches $z > 5$ in images of α Cen B, χ^1 Ori and ϵ Eri, in some images of each star and not in others, and no other strong line of α Cen A (C II, Si IV, C IV, He II, Al II) shows it. The STIS E230H spectrum of α Cen A shows an excess of at most $(1.0 \pm 0.1) \times 10^{-13}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$ within 0.12 Å of 1810.64 Å, where the IUE feature carries 5×10^{-13} to $2 \times 10^{-12}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$, so the feature is not a line of the star but an effect of the SWP camera near 1810 Å.

Of the nine single-channel FUSE peaks, four lie on the Lyβ, C III and O VI lines of Capella, EV Lac and AU Mic, and five reach 5.0 to 5.5σ in AD Leo, Proxima Centauri and HD 340611 with nothing at 3σ in the overlapping channel. None recurs in a second channel.

4.2. Limits

Figure 1 shows the recovery of injected lines and each star's limit, and Table 5 the best limit at each star within 5 pc. Recovery reaches 90% of its plateau at 2.8 (COS), 3.1 (IUE), 3.3 (StarCAT), 3.9 (later STIS) and 4.7 (FUSE) times the nominal threshold, and the plateaus are 0.96, 0.91, 0.90,

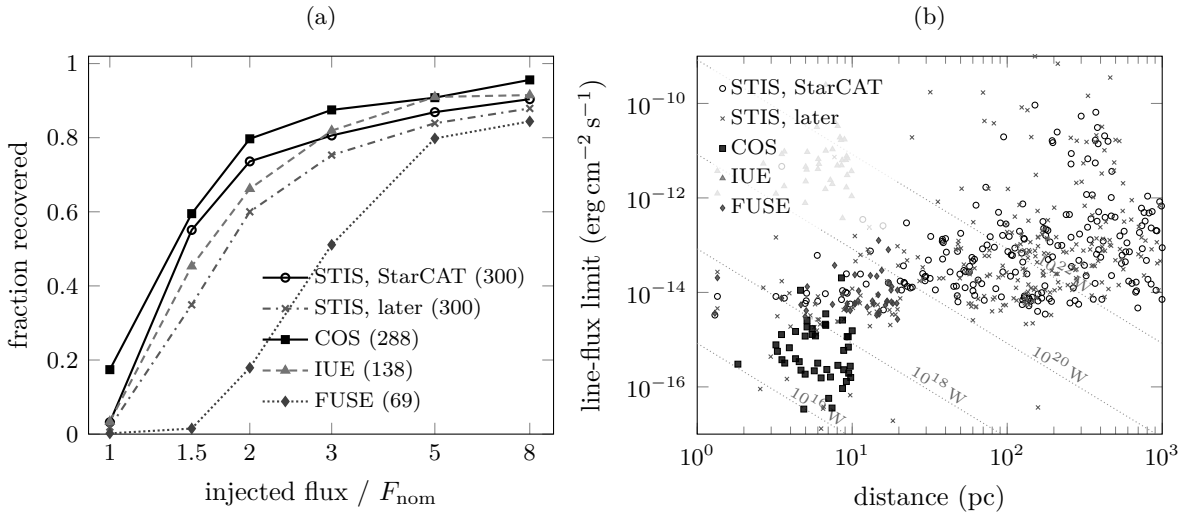


Figure 1: (a) The fraction of injected lines recovered against their flux in units of each exposure's nominal threshold, with the number of exposures in brackets. (b) Each star's limit on an unresolved line against its distance, with dotted lines of constant isotropic luminosity.

0.88 and 0.84. The median limit is 4.8×10^{-16} erg cm^{-2} s^{-1} over the 44 COS stars, 6.3×10^{-15} over the 42 FUSE stars, 4.8×10^{-14} over the 315 StarCAT stars and 4.9×10^{-14} over the 458 later STIS stars with two exposures in one setting, and 5.3×10^{-12} over the 40 IUE stars. At the stars within 10 pc the median limit for an isotropic source is 2.0×10^{17} W with COS, 1.0×10^{18} W with the later STIS spectra of 30 of them, 3.9×10^{18} W with FUSE, 5.1×10^{18} W with StarCAT and 2.9×10^{21} W with IUE, and for a transmitter aimed at Earth through a 10 m aperture 2.7, 13, 34, 95 and 1.4×10^{5} W. Within 5 pc, 21 of the 27 stars are limited to 0.1 to 7 W, Lalande 21185 to 0.11 W, Barnard's star to 0.14 W and Proxima Centauri to 1.0 W, for a beam held on the Earth; a beam pointed at the Sun would miss the Earth by 200 times its width from Barnard's star.

5. Related work

Laser searches. The searches of Section 1 [2, 3, 4, 5, 6] look for unresolved lines in optical spectra from the ground, with thresholds of tens of kilowatts to megawatts for a 10 m transmitter at the distances of their stars; the HARPS search, like this one, finds that the features which survive its tests lie in stars whose own spectra must be excluded first. We find no search of ultraviolet spectra.

Ultraviolet spectra of nearby stars. The archives were assembled to study chromospheres, transition regions, winds and the interstellar medium. StarCAT collects the STIS echelle spectra of 545 stars [10], the pipeline's x1d files hold the later ones, and the COS spectra of nearby M dwarfs come largely from programmes that measure their ultraviolet radiation for the atmospheres of their planets [20]. These are the deepest ultraviolet spectra of the nearest stars, and the narrow emission lines they reveal, such as the Fe II lines that follow Mg II through the pulsation of Mira variables [19], are what an unresolved feature must be told apart from.

Table 5: The best limit at each star within 5 pc. λ is the median wavelength of the spectra that give it, F the line flux at the Earth, $L = 4\pi d^2 F$ the luminosity of an isotropic source and $P_{10\,\mathrm{m}}$ the power of a transmitter aimed at Earth through a diffraction-limited 10 m aperture.

Star	d (pc)	Instrument	λ (Å)	F (erg cm^{-2} s^{-1})	L (W)	$P_{10\,\mathrm{m}}$ (W)
Proxima Centauri	1.30	STIS	1391	3.3×10^{-15}	6.7×10^{16}	1.0×10^{0}
α Cen A	1.35	STIS	1412	8.2×10^{-15}	1.8×10^{17}	2.8×10^{0}
α Cen B	1.35	STIS, later	1384	4.6×10^{-15}	9.9×10^{16}	1.5×10^{0}
Barnard's star	1.83	COS	1212	3.0×10^{-16}	1.2×10^{16}	1.4×10^{-1}
Lalande 21185	2.55	STIS, later	1379	9.0×10^{-17}	7.0×10^{15}	1.1×10^{-1}
Sirius A	2.64	IUE	1563	2.7×10^{-9}	2.3×10^{23}	4.4×10^{6}
Sirius B	2.67	STIS, later	2826	1.3×10^{-14}	1.1×10^{18}	6.8×10^{1}
Ross 154	2.98	STIS, later	1379	4.4×10^{-16}	4.6×10^{16}	7.0×10^{-1}
ϵ Eri	3.22	COS	1372	7.8×10^{-16}	9.6×10^{16}	1.4×10^{0}
GJ 887	3.29	COS	1586	5.7×10^{-16}	7.3×10^{16}	1.5×10^{0}
61 Cyg A	3.50	COS	1362	1.3×10^{-15}	1.9×10^{17}	2.8×10^{0}
Procyon	3.51	STIS, later	1289	1.7×10^{-14}	2.6×10^{18}	3.4×10^{1}
GJ 725 A	3.52	COS	1587	3.8×10^{-16}	5.6×10^{16}	1.1×10^{0}
ϵ Ind	3.64	IUE	2991	1.8×10^{-12}	2.8×10^{20}	2.0×10^{4}
τ Cet	3.65	STIS	1389	7.5×10^{-15}	1.2×10^{18}	1.9×10^{1}
GJ 1061	3.67	COS	1587	3.2×10^{-16}	5.2×10^{16}	1.0×10^{0}
Luyten's star	3.79	STIS, later	1377	1.5×10^{-16}	2.5×10^{16}	3.8×10^{-1}
Kapteyn's star	3.93	COS	1362	6.7×10^{-16}	1.2×10^{17}	1.8×10^{0}
Kruger 60	4.00	STIS, later	1380	1.9×10^{-15}	3.6×10^{17}	5.5×10^{0}
GJ 628	4.31	COS	1586	4.0×10^{-16}	8.8×10^{16}	1.8×10^{0}
van Maanen's star	4.31	COS	1639	1.5×10^{-15}	3.3×10^{17}	7.1×10^{0}
GJ 674	4.55	COS	1217	3.4×10^{-16}	8.5×10^{16}	1.0×10^{0}
LAWD 37	4.64	COS	1668	1.1×10^{-14}	2.9×10^{18}	6.3×10^{1}
GJ 876	4.67	COS	1312	2.3×10^{-16}	5.9×10^{16}	8.1×10^{-1}
DENIS J0255−4700	4.87	COS	1156	3.4×10^{-17}	9.6×10^{15}	1.0×10^{-1}
AD Leo	4.97	COS	1139	1.2×10^{-15}	3.5×10^{17}	3.6×10^{0}
GJ 832	4.97	COS	1312	1.9×10^{-16}	5.5×10^{16}	7.5×10^{-1}

6. Conclusion

No unresolved ultraviolet emission line that a laser aimed at Earth could produce is found in the STIS, COS, IUE and FUSE spectra of nearby stars. Every feature that persists in one star and that the tests against other stars do not remove belongs to the spectrum of an evolved star or is a chromospheric line of ϵ Eri, and at the nearest stars a transmitter of about a watt behind a 10 m aperture, its beam held on the Earth, would have been seen, from 21 of the 27 stars within 5 pc.

The limits apply to lines that are steady or recur in the spectra of a star, that rise above the local upper envelope and that lie away from the masked lines, and to the epochs of the archival spectra. A laser at the wavelength of a strong stellar line, one in an evolved star whose spectrum holds narrow emission lines, or one that was off when the spectra were taken is not constrained. The COS spectra of the stars beyond 10 pc remain to be searched.

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